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Summary of Findings, Conclusions, and Recommendations

7. Summary of Findings

- **Definition**: A concise recap of results obtained from tests, feasibility studies, and analyses.
- **Example**: "The irrigation system achieved a 30% water savings, was technically stable, and had high user acceptance among farmers."
- **Use in Research**: Summarizes evidence to support conclusions.

Summary of Findings

The development of Pandaiku aimed at addressing the challenges commonly face by weavers during their manual drying and pressing of pandan leaves used in weaving purposes. To evaluate the device, the developers was assessed by five (5) CpE Practitioners, five (5) weavers, and one (1) Agriculturist from the Municipality of Luisiana, Laguna. The evaluators were chosen for they have the knowledge regarding the device or the process and functionalities that it covers as well as they can provide meaningful insights that will help in further improvement of the device.

Based on the testing regarding the maintaining of temperature inside the device, the results shows that in fifteen minutes, Pandaiku was able to increase its temperature from 41°C to 57°C which is closer to the set temperature in the PID controller.

On the other hand, three tests has been made to determine the exact time that pandan leaves will be dried. On the first testing with a set temperature of 60°C, it took thirty (30) for a batch of fifty (50) to seventy (70) of pandan leaves to be dried. Another test has been made where the temperature was set to 70°C and it took twenty (20) minutes for the pandan leaves to be dried. While it only took fifteen

8. Conclusions

- **Definition**: Final interpretation of research results, showing whether objectives were achieved.
- **Example**: "The smart irrigation system is technically and economically feasible, and it can significantly improve farming efficiency."
- **Use in Research**: Demonstrates the project's overall success and relevance.

Conclusions

Based on the accuracy of the results from the gathered data, the following conclusions were drawn:

1. The developed device successfully performed the following functions:

    a) Drying Pandan Leaves using Infrared Heating Element;

    b) Monitored processes of drying and pressing using the Thermocouple, PID Controller, and Arduino Mega;

    c) Automatically stop drying based on the present time by powering off the heating element and fans, relays, and Solid State Relay (RRS);

    d) Controlled the pressing mechanism using Arduino-Mega through Solid State Relay (SSR), Wiper Motor, and limit switch;

9. Recommendations

- **Definition**: Suggestions for improvement, future research, or wider application of the project.
- **Example**: "Future work should integrate solar panels for energy independence and expand the system for large-scale farming."
- **Use in Research**: Guides stakeholders, researchers, or future developers on how to build upon the project.

Recommendations

Based on the findings and results of the evaluations and experimentation conducted with the device, the following statements are the recommendations of the project:

1. To encourage the adoption of new way of drying and pressing of pandan leaves, aiming to maximize and increase the production of ready to weave pandan leaves.
2. To implement and integrate an arduino based drying and pressing device to weavers' organization, aiming to increase a better quality of pandan leaves while minimizing manual labor and time consume of the process.